A structure-preserving local discontinuous Galerkin method for the Fokker-Planck-Landau equation

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June 27, 2026

Abstract

In this work, we introduce a structure-preserving local discontinuous Galerkin (LDG) method [Cockburn and Shu(1998)] for solving the nonlocal nonlinear Fokker-Planck-Landau (FPL) equations in both non-relativistic and relativistic regimes. We rephrase the structure-preserving strategy of Shiroto and Sentoku [Shiroto and Sentoku(2019)] in the language of numerical analysis, and extend it to the high-order LDG framework. We propose a method that is not only conservative, but also stabilized through upwind flux. The apparent contradiction between conservation laws and numerical stabilization is elegantly resolved by leveraging the properties of the jump terms inherent to the LDG framework. In the numerical experiments, our scheme is tested with benchmark examples in both low-energy and high-energy regimes.

Keywords: relativistic Landau equation; local discontinuous Galerkin method; structure-preserving method; spline interpolation

1 Introduction

In this work, we consider the Fokker–Planck–Landau (FPL) equation. Developed by Lev Landau in 1936, this model introduced a correction to the Boltzmann equation designed to describe dilute hot plasmas, where charged particles interact via long-range Coulomb forces [Hinton(1983), Lifshitz and Pitaevskii(1981)]. Several variants of the FPL collisional operator have since emerged, by taking the grazing limit of Boltzmann equations for different interaction potentials. In particular, when particle velocities approach the speed of light, relativistic effects become significant, and a relativistic version of the FPL equation was derived by Budker and Beliaev in 1956 [Beliaev and Budker(1956), Belyaev and Budker(1961), Beliaev and Budker(1956)]. Simplified models such as the Dougherty operator [Dougherty(1964)] and the linear Landau operator have also been extensively used in numerical simulations. In general, the FPL collision operator introduces dissipation that drives the system toward thermodynamic equilibrium.

The FPL equation has applications across multiple areas in science and engineering, including controlled fusion, solar wind and radiation belt. In particular, the study of runaway electrons in tokamaks necessitates numerical simulation based on the relativistic version [Dreicer(1959)].

There are several challenges [Dimarco and Pareschi(2014)] associated with numerically solving the Fokker-Planck-Landau (FPL) equation: high dimensionality of the phase space, nonlocality and nonlinearity of the collision operator, the need to preserve physical invariants (mass, momentum, and energy), positivity, numerical stability and the decay of entropy. It is generally impractical to design a single

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numerical method that resolves all of the aforementioned difficulties; practical schemes typically involve compromises tailored to the problem at hand.

A wide variety of deterministic numerical methods have been developed for the FPL equation. Spectral methods [Zhang and Gamba(2017), Pennie and Gamba(2019a), Pennie and Gamba(2019b), Pennie and Gamba(2020), Dimarco et al.(2015)Dimarco, Li, Pareschi and Yan] benefit from low time complexity due to their use of the Fast Fourier Transform and the special convolution structure of the non-relativistic collision operator. The particle method of Carrillo et al. [Carrillo et al.(2020)Carrillo, Hu, Wang and Wu] retains conservation and entropy decay by adopting a regularized collision operator. Hirvijoki et al. [Hirvijoki and Adams(2017)] proposed a conservative FEM/DG scheme using a basis capable of exactly representing second-order polynomials. Daniel et al. [Daniel et al.(2020)Daniel, Taitano and Chacón] proposed an implicit conservative finite difference scheme based on the Fokker–Planck–Rosenbluth formulation. Degond et al. [Degond and Lucquin-Desreux(1994)] used the logarithmic form of the equation to construct an entropy scheme. Other techniques include sublattice multigrid and Monte Carlo methods [Buet et al.(1997)Buet, Cordier, Degond and Lemou], as well as finite volume schemes [Taitano et al.(2015)Taitano, Chacón, Simakov and Molvig]. For the relativistic FPL equation, Shiroto et al. [Shiroto and Sentoku(2019)] proposed a structure-preserving finite difference discretization.

In this paper we propose a local discontinuous Galerkin (LDG) method. Introduced in [Cockburn and Shu(1998)] and further developed in [Cockburn and Dawson(1999), Castillo et al.(2000)Castillo, Cockburn, Perugia and Schötzau, Cockburn et al.(2001)Cockburn, Kanschat, Perugia and Schötzau, Arnold et al.(2002)Arnold, Brezzi, Cockburn and Marini, Cockburn and Shu(2001)], it is well-suited for solving kinetic equations, as it readily supports hp-adaptivity, allows for easy implementation of stabilization techniques, and is naturally amenable to parallelization. Despite preceding works on conservative DG solvers [Shiroto et al.(2022)Shiroto, Matsuyama, Aiba and Yagi, Kim et al.(2022)Kim, Seo, Jo, Kwon and Yoon], our solver is unique in its introduction of *structure-preserving* techniques.

Structure-preserving schemes are conservative, but to be precise, the two concepts are not equivalent. Conservation can be enforced through various means—for instance, by introducing a correction step [Zhang and Gamba(2017), Daniel et al.(2020)Daniel, Taitano and Chacón] that projects the solution back onto the conservative subspace, or by enlarging the test space [Shiroto et al.(2022)Shiroto, Matsuyama, Aiba and Yagi, Hirvijoki and Adams(2017)] to explicitly include momentum and energy. However, such approaches do not necessarily preserve the underlying structure of the collision operator. Specifically, this structure refers to the fact that relative velocity always lies in the null space of the integral kernel, which will be addressed in detail later.

We focus on the structure-preserving method for the relativistic Landau equation since the non-relativistic equation is a low-energy limit of the original one. The paper is organized as follows. Section 2 introduces FPL equations and their properties. In Section 3, we introduce the LDG method and the discrete gradient, along with relevant notations. The structure-preserving strategy is discussed in Section 4. Section 5 presents details associated to implementation. In Section 6, we show numerical results that validate our conservation properties. Finally, Section 7 provides concluding remarks and outlines future directions.

2 The Fokker-Planck-Landau Equations

2.1 General form and properties

The Fokker-Planck-Landau equation has the following structure (we keep the notation consistent with Strain and Taskovic [Strain and Tasković(2019)])

$$\partial_t f = \nabla_p \cdot \left(\int_{\mathbb{R}^3} \Phi(\mathbf{p}, \mathbf{q}) \cdot (f(\mathbf{q}) \nabla_p f(\mathbf{p}) - f(\mathbf{p}) \nabla_q f(\mathbf{q}))\, d\mathbf{q} \right) \quad \text{with} \quad \mathbf{p} \in \mathbb{R}^3 \tag{1}$$

where Φ is a collision kernel determined by the interaction potential. If we let the single particle kinetic energy be $\mathcal{E}(\mathbf{p})$,

$$\mathcal{E}(\mathbf{p}) := \begin{cases} \mathbf{p}^2/2, & \text{non-relativistic,} \\ \sqrt{1+\mathbf{p}^2}, & \text{relativistic.} \end{cases}$$

this collision kernel can be written as the product of a scalar field Λ and a tensor field $\mathbb{S}$:

$$\Phi(\mathbf{p},\mathbf{q}) = \Lambda(\mathbf{p},\mathbf{q})\,\mathbb{S}\left(\nabla_p\mathcal{E}(\mathbf{p}),\nabla_q\mathcal{E}(\mathbf{q})\right).$$

The tensor field $\mathbb{S}(\mathbf{v},\mathbf{w})$ depends on velocities $\mathbf{v},\mathbf{w}\in\mathbb{R}^3$, which are gradients of the kinetic energy, i.e. $\mathbf{v}=\nabla_p\mathcal{E}(\mathbf{p})$ and $\mathbf{w}=\nabla_q\mathcal{E}(\mathbf{q})$. Denoting the relative velocity as $\mathbf{u}:=\mathbf{v}-\mathbf{w}$, and the mean velocity as $\mathbf{z}:=\frac{\mathbf{v}+\mathbf{w}}{2}$, then we consider tensors $\mathbb{S}$ of the following form,

$$\mathbb{S}(\mathbf{v},\mathbf{w}) := \begin{cases}(\mathbf{u}\cdot\mathbf{u})\mathbb{I}-\mathbf{u}\otimes\mathbf{u}, & \text{non-relativistic,}\\ (\mathbf{u}\cdot\mathbf{u})\mathbb{I}-\mathbf{u}\otimes\mathbf{u}-(\mathbf{z}\times\mathbf{u})\otimes(\mathbf{z}\times\mathbf{u}), & \text{relativistic.}\end{cases} \tag{2}$$

This tensor is the key to energy conservation and entropy decay. Indeed, by testing against a suitable test function $\varphi(\mathbf{p})$ we obtain the weak form,

$$\int_{\mathbb{R}^3}\varphi\partial_t f\,d\mathbf{p} = -\int_{\mathbb{R}^3}\int_{\mathbb{R}^3}\Phi(\mathbf{p},\mathbf{q}):\left[\nabla_p\varphi(\mathbf{p})\otimes\left(f(\mathbf{q})\nabla_p f(\mathbf{p})-f(\mathbf{p})\nabla_q f(\mathbf{q})\right)\right]\,d\mathbf{q}\,d\mathbf{p}. \tag{3}$$

Since the collision kernel is symmetric, i.e. $\Phi(\mathbf{p},\mathbf{q})=\Phi(\mathbf{q},\mathbf{p})$, it can be rewritten as:

$$\int_{\mathbb{R}^3}\varphi\partial_t f\,d\mathbf{p} = -\int_{\mathbb{R}^3}\int_{\mathbb{R}^3}\Phi(\mathbf{p},\mathbf{q}):\left[\left(\nabla_q\varphi(\mathbf{q})-\nabla_p\varphi(\mathbf{p})\right)\otimes f(\mathbf{p})\nabla_q f(\mathbf{q})\right]\,d\mathbf{q}\,d\mathbf{p}. \tag{4}$$

Based on Equation (4), we are able to prove the following conservation laws.

Theorem 1 (Conservation Laws)**.** *Suppose that $f(\mathbf{p},t)$ is the solution to the FPL equation* (1)*, then we have*

$$\partial_t(f,1)=0$$
$$\partial_t(f,\mathbf{p})=0$$
$$\partial_t(f,\mathcal{E}(\mathbf{p}))=0$$

Proof. One can immediately observe that the mass $\int_{\mathbb{R}^3}f\ d\mathbf{p}$ and momentum $\int_{\mathbb{R}^3}f\mathbf{p}\ d\mathbf{p}$ are conserved since $\nabla_q\varphi(\mathbf{q})-\nabla_p\varphi(\mathbf{p})=0$ when $\varphi(\mathbf{p})=1$ or $\mathbf{p}$.

As for energy conservation, note that the vectors $\mathbf{u}$ and $\mathbf{z}$ generate a set of orthogonal basis vectors: $\mathbf{u}$, $\mathbf{z}\times\mathbf{u}$ and $\mathbf{u}\times(\mathbf{z}\times\mathbf{u})$. They are, in fact, three eigenvectors of the tensor $\mathbb{S}$ in (2). In particular, the relative velocity $\mathbf{u}$ is an eigenvector whose eigenvalue is zero, i.e.

$$\mathbb{S}\cdot\mathbf{u}=\mathbb{S}\left(\nabla_p\mathcal{E}(\mathbf{p}),\nabla_q\mathcal{E}(\mathbf{q})\right)\cdot\left(\nabla_p\mathcal{E}(\mathbf{p})-\nabla_q\mathcal{E}(\mathbf{q})\right)=0.$$

As a consequence, the FPL equation always preserves energy $\int_{\mathbb{R}^3}f\mathcal{E}\ d\mathbf{p}$. □

Remark 1. *In classical mechanics, the velocity of a particle is proportional to its momentum, while in the theory of relativity it is not. To be as general as possible, we use $\nabla_p\mathcal{E}(\mathbf{p})$ to represent velocity. And as we will see later, this relation between velocity and kinetic energy is the key to a structure-preserving strategy.*

Moreover, the positive semi-definiteness of tensor $\mathbb{S}$ leads to entropy dissipation

$$\partial_t\left(\int_{\mathbb{R}^3}f\log f\ d\mathbf{p}\right)\leqslant 0.$$

A key step in designing our method is distributing the tensor-vector product in the strong form (1) in order to obtain a *nonlinear, nonlocal, advection-diffusion equation*

$$\partial_t f=\nabla_p\cdot\left(\mathcal{D}\nabla_p f-\mathcal{U}f\right), \tag{5}$$

where

$$\mathcal{U}(\mathbf{p})=\int_{\mathbb{R}^3}\Phi(\mathbf{p},\mathbf{q})\nabla_q f(\mathbf{q})\,d\mathbf{q}, \tag{6}$$

and

$$\mathcal{D}(\mathbf{p})=\int_{\mathbb{R}^3}\Phi(\mathbf{p},\mathbf{q})f(\mathbf{q})\,d\mathbf{q}. \tag{7}$$

Note that the positive semi-definiteness of tensor field $\mathbb{S}$ and the non-negativity of the density f guarantee the positive semi-definiteness of diffusion coefficient $\mathcal{D}(\mathbf{p})$.

3 The Local Discontinuous Galerkin Discretization

Given a probability density function $f(\mathbf{p})$, there can be subsets of the momentum space $\mathbb{R}^3$, where advection is dominant, i.e. $\|\mathcal{U}\| \gg \|\mathcal{D}\|$. Moreover, when coupled with the Vlasov equation, the magnitude of the Lorentz force can easily exceed that of the collision term. From this perspective, stabilization techniques such as upwinding become necessary, which motivates the use of the local discontinuous Galerkin (LDG) method.

In this section we will define the LDG method for an advection-diffusion equation in semi-discrete form. Note that the diffusion coefficient $\mathcal{D}$ is a tensor rather than a constant, therefore we take the L^2-projection of the diffusion flux, as was proposed in [Cockburn and Dawson(1999)]. We will also introduce the concept of *discrete gradient operator*, which is the key ingredient to our structure-preserving scheme.

3.1 Notations, definitions and projections

As has been proved in [Guillen and Silvestre(2025)], the solution of a homogeneous FPL equation is always bounded by a Gaussian distribution determined by the initial condition only. Therefore given any $0 < \epsilon \ll 1$, there exists a finite domain $\Omega \subsetneq \mathbb{R}^3$ such that for any $t \geqslant 0$,

$$\left| 1 - \frac{\int_\Omega f(\mathbf{p}, t) d\mathbf{p}}{\int_{\mathbb{R}^3} f(\mathbf{p}, t) d\mathbf{p}} \right| \leqslant \epsilon.$$

Hence it is reasonable to assume a compact support for the solution and truncate the computational domain, analogous to existing work on kinetic equations, for example [Zhang and Gamba(2017), Zhang and Gamba(2018)].

To solve on a cut-off domain, a suitable boundary condition is necessary. As we know, on the boundary $\partial\Omega$ of a large enough cut-off domain Ω, both the value of the density and the flux across the boundary are nearly zero. Therefore the numerical solution is indifferent to the choice of boundary condition, as long as it is homogeneous, i.e. one that admits the trivial solution, zero. We are free to choose one that simplifies the implementation and analysis later on.

Let $\mathbf{n}$ denote the unit outward normal to $\Gamma \equiv \partial\Omega$.

In the rest of this paper, we will use the following zero-flux Robin condition on the boundary:

$$(\mathcal{U} f - \mathcal{D} \cdot \nabla_{\mathbf{p}} f) \cdot \mathbf{n} = 0, \text{ on } \Gamma, \tag{8}$$

We start motivating the LDG method by rewriting (5) in the following mixed form:

$$\partial_t f = \nabla_{\mathbf{p}} \cdot (Z - \mathcal{U} f).$$
$$Z = \mathcal{D} \cdot \widetilde{Z}$$

with

$$\widetilde{Z} = \nabla_{\mathbf{p}} f. \tag{9}$$

Let the test space be $\mathcal{W}$, assuming smooth enough solutions, we get the following weak formulation of the equations above. Denote the volume integral as $()$ and denote the trace integral as $\langle\rangle$,

$$(\partial_t f, \varphi)_\Omega = (\mathcal{U} f - Z, \nabla_p \varphi)_\Omega, \quad \forall\, \varphi \in \mathcal{W}. \tag{10a}$$

$$\left(Z, \widetilde{V}\right)_\Omega = \left(\mathcal{D}\widetilde{Z}, \widetilde{V}\right)_\Omega, \quad \forall\, \widetilde{V} \in (\mathcal{W})^3. \tag{10b}$$

$$\left(\widetilde{Z}, V\right)_\Omega = -\left(f, \nabla_{\mathbf{p}} \cdot V\right)_\Omega + \langle f, V^- \cdot \mathbf{n}^- \rangle_{\partial\Omega}, \quad \forall\, V \in (\mathcal{W})^3. \tag{10c}$$

Let $\mathcal{T}_h = \{R\}$ be a partition of Ω, with R being Cartesian elements or simplices. Denote the mesh size of $\mathcal{T}_h$ as $h = \max_{R \in \mathcal{T}_h} \operatorname{diam}(R)$. Furthermore the inner products inside each element R are defined as

$$(g, h)_R = \int_R g h \, d\mathbf{p}.$$

Let $\mathcal{I}_h$ be the set of element faces, the integral on each face e is denoted as

$$\langle g, h\rangle_e = \int_e gh\, d\mathbf{s}.$$

For piecewise functions defined with respect to $\mathcal{T}_h$, we introduce the jumps and averages as follows. For any interface $e = \{R^+ \cap R^-\} \in \mathcal{I}_h$, with $\mathbf{n}^\pm$ as the outward unit normal to $\partial R^\pm$, $g^\pm = g|_{R^\pm}$, the jump across e is defined as

$$[g] = g^+\mathbf{n}^+ + g^-\mathbf{n}^-. \tag{11}$$

and the average across e is

$$\{g\} = \frac{1}{2}(g^+ + g^-). \tag{12}$$

Next we define the discrete test space

$$\mathcal{W}_h = \left\{g \in L^2(\Omega) : g|_R \in \mathbb{Q}^k(R), \forall R \in \mathcal{T}_h\right\}. \tag{13}$$

where $\mathbb{Q}^k(R)$ is the space of polynomials of degree at most k in each variable.

By replacing the infinite dimensional test space $\mathcal{W}$ with the finite dimensional space $\mathcal{W}_h$, we obtain the semi-discrete weak form as follows: find $\partial_t f_h \in \mathcal{W}_h, Z_h \in \mathcal{W}_h^3, \widetilde{Z}_h \in \mathcal{W}_h^3$ such that

$$\begin{aligned}\sum_R (\partial_t f_h, \varphi_h)_R - \sum_R (\mathcal{U} f_h, \nabla_p \varphi_h)_R + \sum_R \langle \widehat{\widehat{\mathcal{U} f_h}}, \varphi_h^- \mathbf{n}^-\rangle_{\partial R/\Gamma} \\ = -\sum_R (Z_h, \nabla\varphi_h)_R + \sum_R \langle \widehat{Z}_h, \varphi_h^- \mathbf{n}^-\rangle_{\partial R/\Gamma}, \forall\ \varphi_h \in \mathcal{W}_h,\end{aligned} \tag{14a}$$

$$\sum_R \left(Z_h, \widetilde{V}_h\right)_R = \sum_R \left(\mathcal{D}\cdot\widetilde{Z}_h, \widetilde{V}_h\right)_R, \forall\ \widetilde{V}_h \in \mathcal{W}_h^3, \tag{14b}$$

$$\sum_R \left(\widetilde{Z}_h, V_h\right)_R = -\sum_R (f_h, \nabla\cdot V_h)_R + \sum_R \langle \check{f}_h, V_h^- \cdot \mathbf{n}^-\rangle_{\partial R}, \forall\ V_h \in \mathcal{W}_h^3. \tag{14c}$$

Since the discrete solutions $f_h, Z_h, \widetilde{Z}_h$ are discontinuous on the interfaces, the flux terms $\widehat{\widehat{\mathcal{U} f_h}}, \widehat{Z}_h, \check{f}_h$ remain to be determined. They will be addressed later in (15), (16), and (28).

3.2 The bilinear form of the LDG method

In the previous subsection, we introduced the LDG method, and left the key ingredient, flux terms, undetermined. Our conservation strategy is only possible with alternating flux. Before demonstrating the reason, we have to give a brief introduction on some prerequisite knowledge in this subsection.

In what follows, we will introduce a key concept: the *discrete gradient operator* ∇_h, and show that alternating flux enables a special bilinear form of LDG.

Choosing an appropriate reference direction $\mathbf{u}$, say $\mathbf{u} = (1, 1, 1)$, we use alternating flux as follows:

$$\widehat{Z} = \begin{cases} Z^-, & \mathbf{n}^-\cdot\mathbf{u} > 0 \\ Z^+, & \text{otherwise}\end{cases} \tag{15}$$

and

$$\check{f} = \begin{cases} f^-, & \mathbf{n}^-\cdot\mathbf{u} < 0 \text{ or } \partial R \subset \partial\Omega \\ f^+, & \text{otherwise}\end{cases} \tag{16}$$

Now recall the definition of $\widetilde{Z}$ in Equation (10c), it is just an L^2-projection of $\nabla_p f$ into the test space $\mathcal{W}_h$. In other words, $\widetilde{Z}$ is a discrete gradient of f. The map from f to its discrete gradient $\widetilde{Z}$ is called the discrete gradient operator.

Definition 1 (discrete gradient operator)**.** *We define $\nabla_h f_h$ as the function in $\mathcal{W}_h$ such that,*

$$\sum_R (\nabla_h f_h, V_h)_R = -\sum_R (f_h, \nabla \cdot V_h)_R + \sum_R \langle \widecheck{f}_h, V_h^- \cdot \mathbf{n}^- \rangle_{\partial R}, \forall V_h \in \mathcal{W}_h^3. \tag{17}$$

The operator ∇_h is the discrete gradient operator with respect to the flux $\widecheck{f}$ in (16)*.*

It is natural to ask what the difference is between a gradient operator and a discrete gradient operator. The following proposition shows that for discontinuous piecewise polynomial f, the only difference comes from the jumps on the interfaces.

Proposition 1. *If $f_h \in \mathcal{W}_h$, then $\nabla_h f_h$ satisfies that*

$$-\sum_R (\nabla_h f_h, V_h)_R = -\sum_R (\nabla f_h, V_h)_R + \sum_R \langle f_h^- \mathbf{n}^-, \widehat{V_h} \rangle_{\partial R/\Gamma}, \forall V_h \in \mathcal{W}_h^3 \tag{18}$$

where $\widehat{V_h}$ is the alternating flux in (15)*.*

Consequently, if we define the lift operator r as in [Arnold et al.(2002)Arnold, Brezzi, Cockburn and Marini]:

$$\sum_R (r([[f_h]]), V_h)_R = -\sum_{l_e \in \partial R/\Gamma} \langle [[f_h]], \widehat{V_h} \rangle_{l_e}, \forall V_h \in \mathcal{W}_h^3,$$

then

$$\nabla_h f_h = \nabla f_h + r([[f_h]]).$$

Proof. Integrating by parts on the right-hand side of (17), we have

$$\begin{aligned}\sum_R (\nabla_h f_h, V_h)_R &= -\sum_R (f_h, \nabla \cdot V_h)_R + \sum_R \langle \widecheck{f_h}, V_h^- \cdot \mathbf{n}^- \rangle_{\partial R} \\ &= \sum_R (\nabla f_h, V_h)_R + \sum_R \langle \widecheck{f}_h - f_h^-, V_h^- \cdot \mathbf{n}^- \rangle_{\partial R/\Gamma}\end{aligned}$$

Comparing with (18), it remains to show that

$$\sum_R \langle \widecheck{f}_h - f_h^-, V_h^- \cdot \mathbf{n}^- \rangle_{\partial R/\Gamma} = -\sum_R \langle f_h^- \mathbf{n}^-, \widehat{V_h} \rangle_{\partial R/\Gamma}.$$

Note that in the above summation, each interface is counted twice. Suppose that on the interface between R_l and R_r we have $\widecheck{f_h} = f_h^l$ and $\widehat{V_h} = V_h^r$, then

$$\begin{aligned}&\sum_{R_l, R_r} \langle \widecheck{f_h}, V_h^- \cdot \mathbf{n}^- \rangle_{\partial R_l \cap \partial R_r} + \sum_{R_l, R_r} \langle f_h^- \mathbf{n}^-, \widehat{V_h} \rangle_{\partial R_l \cap \partial R_r} \\ =& \langle f_h^l, V_h^l \cdot \mathbf{n}^l \rangle_{\partial R_l \cap \partial R_r} + \langle f_h^l, V_h^r \cdot \mathbf{n}^r \rangle_{\partial R_l \cap \partial R_r} \\ &+ \langle f_h^l \mathbf{n}^l, V_h^r \rangle_{\partial R_l \cap \partial R_r} + \langle f_h^r \mathbf{n}^r, V_h^r \rangle_{\partial R_l \cap \partial R_r} \\ =& \langle f_h^l, V_h^l \cdot \mathbf{n}^l \rangle_{\partial R_l \cap \partial R_r} + \langle f_h^r \mathbf{n}^r, V_h^r \rangle_{\partial R_l \cap \partial R_r}\end{aligned}$$

It follows that

$$\sum_R \langle \widecheck{f_h}, V_h^- \cdot \mathbf{n}^- \rangle_{\partial R/\Gamma} + \sum_R \langle f_h^- \mathbf{n}^-, \widehat{V_h} \rangle_{\partial R/\Gamma} = \sum_R \langle f_h^-, V_h^- \cdot \mathbf{n}^- \rangle_{\partial R/\Gamma}$$

$\square$

With the discrete gradient operator, we will be able to rewrite the diffusion part in a compact bilinear form [Arnold et al.(2002)Arnold, Brezzi, Cockburn and Marini], without involving the auxiliary field $\widetilde{Z}$ as in (9).

Theorem 2. *The whole system (14) is equivalent to*

$$(\partial_t f_h, \varphi_h)_\Omega + (\nabla_h f_h, \mathcal{D} \cdot \nabla_h \varphi_h)_\Omega = (\mathcal{U} f_h, \nabla \varphi_h)_\Omega - \sum_R \langle \widehat{\mathcal{U} f_h}, \varphi_h^- \mathbf{n}^- \rangle_{\partial R/\Gamma} \tag{19}$$

Proof. To shorten the length of the formulas, we introduce the following notation:

$$\mathcal{A} = (\mathcal{U} f_h, \nabla \varphi_h)_\Omega - \sum_R \langle \widehat{\mathcal{U} f_h}, \varphi_h^- \mathbf{n}^- \rangle_{\partial R/\Gamma}$$

Recall the first row of (14). By Proposition 1 we have:

$$\sum_R (\partial_t f_h, \varphi_h)_R = -\sum_R (\nabla_h \varphi_h, Z_h)_R + \mathcal{A}.$$

Since $\nabla_h \varphi \in \mathcal{W}_h$, it follows from Equation (14b) that

$$\sum_R (\partial_t f_h, \varphi_h)_R = -\sum_R \left(\mathcal{D} \cdot \widetilde{Z}_h, \nabla_h \varphi_h\right)_R + \mathcal{A}.$$

By definition, the last row of (14) is equivalent to

$$\widetilde{Z}_h = \nabla_h f_h$$

Therefore, the diffusion part can be written as a bilinear form, and we have

$$\sum_R (\partial_t f_h, \varphi_h)_R = -\sum_R (\mathcal{D} \cdot \nabla_h f_h, \nabla_h \varphi_h)_R + \mathcal{A}.$$

□

4 Structure-Preserving Schemes

In this section, we first introduce the concept of variational crime, as it helps us understand the structure-preserving strategy. We then demonstrate how this strategy generally conflicts with stabilization, and finally propose an upwind structure-preserving LDG scheme.

4.1 Variational Crime

A *variational crime* [Strang(1972)] occurs when the discrete problem does not strictly follow the form of the continuous variational problem. For example, when solving the simple elliptic PDE

$$\nabla \cdot (\mathcal{D} \cdot \nabla f) = \sigma$$

with finite element method, the standard approach is to solve the Galerkin variational problem: find $f_h \in \mathcal{W}_h$ such that

$$B(f_h, \varphi_h) := \int_\Omega (\mathcal{D} \cdot \nabla f_h) \cdot \nabla \varphi_h = S(\varphi) := \int_\Omega \sigma \varphi_h, \forall \varphi_h \in \mathcal{W}_h.$$

However, in practice, it may be impossible to calculate the integral $B(f_h, \varphi_h)$ exactly. One must then approximate the integral by numerical quadrature, approximate the diffusion coefficient $\mathcal{D}$ by piecewise polynomials to obtain an exact integral, or make a similar compromise.

Despite the "crime", many such approximations still yield convergence, but they must be analyzed carefully.

In solving the FPL equation, variational crimes are usually unavoidable due to the complexity of the collision kernel $\Phi(\mathbf{p}, \mathbf{q})$, which makes exact integration infeasible and necessitates the use of quadrature. Since such variational crimes are inevitable, it is reasonable to take advantage of them to serve other purposes.

It turns out that a structure-preserving scheme can be obtained by replacing the collision kernel Φ with a specifically designed approximation Φ_h.

4.2 Symmetric Structure-Preserving Scheme

In [Shiroto and Sentoku(2019)], Shiroto and Sentoku proposed a structure-preserving strategy for finite difference schemes which renders rigorous conservation laws. As we know, a finite difference scheme is equivalent to a discontinuous Galerkin scheme with piecewise constant basis functions. In this subsection, we generalize that strategy to LDG with higher order polynomials.

The conservation laws of FPL equations follow from a special structure in their weak forms. Recall the weak form in Equation (4)

$$\int_{\mathbb{R}^3} \varphi \partial_t f \, d\mathbf{p} = -\int_{\mathbb{R}^3}\int_{\mathbb{R}^3} \Lambda(\mathbf{p},\mathbf{q})\mathbb{S}(\mathbf{v},\mathbf{w}) : [(\nabla_q \varphi(\mathbf{q}) - \nabla_p \varphi(\mathbf{p})) \otimes f(\mathbf{p})\nabla_q f(\mathbf{q})] \, d\mathbf{q}\, d\mathbf{p}.$$

Energy conservation relies on the fact that

$$\mathbb{S}\left(\nabla_p \mathcal{E}(\mathbf{p}), \nabla_q \mathcal{E}(\mathbf{q})\right) \cdot \left(\nabla_p \mathcal{E}(\mathbf{p}) - \nabla_q \mathcal{E}(\mathbf{q})\right) = 0.$$

Considering that the single particle kinetic energy $\mathcal{E}(\mathbf{p})$ may not belong to the test function space $\mathcal{W}_h$, we pursue a numerical scheme that conserves discrete energy, i.e. a scheme such that

$$\partial_t \left(f_h, \Pi_h \mathcal{E}\right)_\Omega = 0$$

where Π_h is a projection to the test space $\mathcal{W}_h$ (which will be specified in subsection 5.1), such that

$$\lim_{h\to 0} \|\mathcal{E} - \Pi_h \mathcal{E}\|_{L^2(\Omega)} = 0 \tag{20}$$

$$\lim_{h\to 0, r(\Omega)\to\infty} \|\nabla \mathcal{E} - \nabla_h \Pi_h \mathcal{E}\|_{L^2(\Omega;\mu)} = 0 \tag{21}$$

where

$$\|\mathbf{F}\|^2_{L^2(\Omega;\mu)} = \int_\Omega \mathbf{F}(\mathbf{p}) \cdot \mathbf{F}(\mathbf{p})\, \mu(\mathbf{p})\, d\mathbf{p}. \tag{22}$$

Here $\mu = \exp(-\mathbf{p}\cdot\mathbf{p}/2r(\Omega)^2)$, where $r(\Omega)$ denotes the *diameter* of Ω.

Remark 2. *Condition* (21) *is important since we expect the error at the tails of the numerical distribution to be negligible, i.e., at machine precision.*

The following scheme is a generalization of the one developed by [Shiroto and Sentoku(2019)],

Theorem 3 (Symmetric Structure-Preserving Scheme)**.** *If $f_h(\mathbf{p},t)$ is a solution to the following semi-discrete weak form,*

$$\int_\Omega \varphi_h \partial_t f_h \, d\mathbf{p} = -\int_\Omega\int_\Omega \Phi_h(\mathbf{p},\mathbf{q}) : [(\nabla_{h,q}\varphi_h(\mathbf{q}) - \nabla_{h,p}\varphi_h(\mathbf{p})) \otimes f_h(\mathbf{p})\nabla_{h,q} f_h(\mathbf{q})] \, d\mathbf{q}\, d\mathbf{p}, \tag{23}$$

where $\nabla_{h,p}$ and $\nabla_{h,q}$ are the discrete gradient operator defined in (17) *and the discrete collision kernel is defined as*

$$\Phi_h(\mathbf{p},\mathbf{q}) = \Lambda\left(\mathbf{p},\mathbf{q}\right)\mathbb{S}\left(\nabla_{h,p}\Pi_h\mathcal{E}(\mathbf{p}), \nabla_{h,q}\Pi_h\mathcal{E}(\mathbf{q})\right), \tag{24}$$

then the following discrete conservation laws hold:

$$\partial_t \left(f_h, 1\right)_\Omega = 0$$
$$\partial_t \left(f_h, \Pi_h \mathbf{p}\right)_\Omega = 0$$
$$\partial_t \left(f_h, \Pi_h \mathcal{E}(\mathbf{p})\right)_\Omega = 0$$

Proof. Mass and momentum conservation laws are trivial. For energy conservation, note that when substituting φ_h with $\Pi_h\mathcal{E}(\mathbf{p})$ in (23), the integrand on the right-hand side is proportional to the following term:

$$\mathbb{S}\left(\nabla_{h,p}\Pi_h\mathcal{E}(\mathbf{p}), \nabla_{h,q}\Pi_h\mathcal{E}(\mathbf{q})\right) \cdot \left(\nabla_{h,p}\Pi_h\mathcal{E}(\mathbf{p}) - \nabla_{h,q}\Pi_h\mathcal{E}(\mathbf{q})\right),$$

which is always equal to zero since for any $\mathbf{v},\mathbf{w} \in \mathbb{R}^3$,

$$\mathbb{S}\left(\mathbf{v},\mathbf{w}\right) \cdot \left(\mathbf{v} - \mathbf{w}\right) = 0.$$

□

Note that the strategy does not rely on the choice of basis functions. For certain choices, it reproduces existing schemes:

- For non-relativistic FPL equation, if all basis functions are continuous and the test space $\mathcal{W}_h$ contains the kinetic energy $\mathcal{E}(\mathbf{p}) = |\mathbf{p}|^2/2$, then $\nabla_{h,p}\Pi_h\mathcal{E}(\mathbf{p}) = \nabla_p\mathcal{E}(\mathbf{p})$, and (23) degenerates to a naive discretization as follows:

$$\int_\Omega \varphi_h \partial_t f_h \, d\mathbf{p} = -\int_\Omega \int_\Omega \Phi(\mathbf{p},\mathbf{q}) : [(\nabla_q\varphi_h(\mathbf{q}) - \nabla_p\varphi_h(\mathbf{p})) \otimes f_h(\mathbf{p})\nabla_q f_h(\mathbf{q})] \, d\mathbf{q}\, d\mathbf{p}.$$

- For relativistic FPL equation, if all basis functions are continuous, then (23) recovers the finite element scheme proposed in [Hirvijoki(2019)].

- For relativistic FPL equation, if we use piecewise constant basis functions on a rectangular mesh in Cartesian coordinate system, then the discrete gradient operator can be explicitly written as

$$\nabla_{h,p}\varphi_h(\mathbf{p}) = \left[\frac{\varphi_h^{i+1,j,k} - \varphi_h^{i,j,k}}{\Delta p}, \frac{\varphi_h^{i,j+1,k} - \varphi_h^{i,j,k}}{\Delta p}, \frac{\varphi_h^{i,j,k+1} - \varphi_h^{i,j,k}}{\Delta p}\right]$$

  where Δp is the mesh size and i, j, k are the mesh indices. Our scheme (23) is then equivalent to the finite difference scheme proposed in [Shiroto and Sentoku(2019)].

Remark 3. *The same strategy not only applies to the FPL equation, but also to the quasilinear theory for particle-wave interaction [Huang et al.(2023)Huang, Abdelmalik, Breizman and Gamba, Huang et al.(2026)Huang, Gamba and Shu].*

4.3 Upwind Structure-Preserving Scheme

As shown in Section 2, the FPL equation can be regarded as a nonlinear nonlocal advection-diffusion equation. It is well-known that advection terms can cause numerical instability or sub-optimal convergence rate, unless the scheme is stabilized.

Let us analyze the symmetric structure-preserving scheme from this new perspective. Note that (23) can be rephrased as follows:

$$\sum_R (\partial_t f_h, \varphi_h)_R + \sum_R (\nabla_{h,p} f_h, \mathcal{D}_h \cdot \nabla_{h,p}\varphi_h)_R = \sum_R (\mathcal{U}_h f_h, \nabla_{h,p}\varphi_h)_R \tag{25}$$

where $\mathcal{D}_h(\mathbf{p}) = \int_\Omega \Phi_h(\mathbf{p},\mathbf{q}) f_h(\mathbf{q})\, d\mathbf{q}$ and $\mathcal{U}_h(\mathbf{p}) = \int_\Omega \Phi_h(\mathbf{p},\mathbf{q}) \cdot \nabla_{h,q} f_h(\mathbf{q})\, d\mathbf{q}$.

Recall that the discrete gradient operator can be expressed as follows,

$$\nabla_h\varphi_h = \nabla\varphi_h + r([[\varphi_h]]),$$

therefore the advection part in (25) is apparently not upwind. Nevertheless, comparing it with the upwind scheme in (19), the only difference comes from $[[\varphi_h]]$, the jump of the test function across element interfaces. That means if mass, momentum and energy are continuous functions on Ω, then upwind will not interfere with conservation. Based on this idea, we propose the following upwind structure-preserving method.

Theorem 4 (Upwind Structure-Preserving Scheme)**.** *Suppose there exists a non-zero projection Π_h into $\mathcal{W}_h \cap C^1(\Omega)$. Let $f_h(\mathbf{p}, t)$ be a solution to the following semi-discrete weak form,*

$$\sum_R (\partial_t f_h, \varphi_h)_R + \sum_R (\nabla_h f_h, \mathcal{D}_h \cdot \nabla_h\varphi_h)_R = \sum_R (\mathcal{U}_h f_h, \nabla\varphi_h)_R - \sum_R \langle \widehat{\mathcal{U}_h f_h}, \varphi_h^- \mathbf{n}^- \rangle_{\partial R/\Gamma} \tag{26}$$

where $\mathcal{D}_h(\mathbf{p}) = \int_\Omega \Phi_h(\mathbf{p},\mathbf{q}) f_h(\mathbf{q}) d\mathbf{q}$ and $\mathcal{U}_h(\mathbf{p}) = \int_\Omega \Phi_h(\mathbf{p},\mathbf{q}) \cdot \nabla_{h,q} f_h(\mathbf{q}) d\mathbf{q}$ and the kernel Φ_h is defined as in Equation (24). Then the following discrete conservation laws hold:

$$\partial_t (f_h, \Pi_h 1)_\Omega = 0$$
$$\partial_t (f_h, \Pi_h \mathbf{p})_\Omega = 0$$
$$\partial_t (f_h, \Pi_h \mathcal{E}(\mathbf{p}))_\Omega = 0.$$

Moreover, the L^2 stability is enhanced as follows:

$$
\begin{aligned}
&\frac{1}{2}\partial_t \left(f_h, f_h\right)_\Omega + \left(\nabla_h f_h, \mathcal{D}_h \cdot \nabla_h f_h\right)_\Omega + \left(\frac{1}{2}\nabla \cdot \mathcal{U}_h, f_h^2\right)_\Omega - \langle \frac{1}{2}\mathcal{U}_h \cdot \mathbf{n}^-, (f_h^-)^2 \rangle_\Gamma \\
=& - \sum_{l_e \in \partial R/\Gamma} \langle \frac{|\mathcal{U}_h \cdot \mathbf{n}|}{2} [[f_h]], [[f_h]] \rangle_{l_e} \\
\leqslant& 0.
\end{aligned}
\tag{27}
$$

Proof. Since the discrete energy $\Pi_h \mathcal{E}(\mathbf{p})$ belongs to $\mathcal{W}_h \cap C^1(\Omega)$, its discrete gradient must be continuous, i.e. $\nabla_{h,p}\Pi_h\mathcal{E}(\mathbf{p}) \in C^0(\Omega)$. Consequently the tensor field $\mathbb{S}\left(\nabla_{h,p}\Pi_h\mathcal{E}(\mathbf{p}), \nabla_{h,q}\Pi_h\mathcal{E}(\mathbf{q})\right)$, as a composition of continuous functions, is continuous in both $\mathbf{p}$ and $\mathbf{q}$. Recall the definition of advection field $\mathcal{U}_h(\mathbf{p})$:

$$
\mathcal{U}_h(\mathbf{p}) = \int_\Omega \Phi_h(\mathbf{p}, \mathbf{q}) \cdot \nabla_{h,q} f_h(\mathbf{q}) d\mathbf{q} = \int_\Omega \Lambda(\mathbf{p}, \mathbf{q}) \mathbb{S}\left(\nabla_{h,p}\Pi_h\mathcal{E}(\mathbf{p}), \nabla_{h,q}\Pi_h\mathcal{E}(\mathbf{q})\right) \cdot \nabla_{h,q} f_h(\mathbf{q}) d\mathbf{q}
$$

It follows that $\mathcal{U}_h(\mathbf{p}) \in C^0(\Omega)$. Hence the upwind flux

$$
\widehat{\mathcal{U}_h f_h} \cdot \mathbf{n}^- = \begin{cases} \mathcal{U}_h f_h^- \cdot \mathbf{n}^-, & \mathcal{U}_h \cdot \mathbf{n}^- > 0 \\ \mathcal{U}_h f_h^+ \cdot \mathbf{n}^-, & \mathcal{U}_h \cdot \mathbf{n}^- < 0 \end{cases}
\tag{28}
$$

is well-defined. Substitute φ_h with f_h in Equation (26), we obtain Equation (27).

Now let us consider the conservation laws. Recall that for any test function $\varphi_h \in \mathcal{W}_h$, its discrete gradient consists of the interior gradient part and the interface jump part:

$$
\nabla_h \varphi_h = \nabla \varphi_h + r([[\varphi_h]])
\tag{29}
$$

Hence the semi-discrete form (26) is equivalent to

$$
\begin{aligned}
\partial_t \left(f_h, \varphi_h\right)_\Omega =& - \int_\Omega \int_\Omega \Phi_h(\mathbf{p}, \mathbf{q}) : \left[\left(\nabla_{h,q}\varphi_h(\mathbf{q}) - \nabla_{h,p}\varphi_h(\mathbf{p})\right) \otimes f_h(\mathbf{p}) \nabla_{h,q} f_h(\mathbf{q}) \right] d\mathbf{q} d\mathbf{p} \\
&+ \left\{ \left(\mathcal{U}_h f_h, r([[\varphi_h]])\right)_\Omega - \sum_{l_e \in \partial R/\Gamma} \langle \mathcal{U}_h \{f_h\}, [[\varphi_h]] \rangle_{l_e} - \sum_{l_e \in \partial R/\Gamma} \langle \frac{|\mathcal{U}_h \cdot \mathbf{n}|}{2} [[f_h]], [[\varphi_h]] \rangle_{l_e} \right\}
\end{aligned}
\tag{30}
$$

As have been proved in Theorem 3, the right hand side of the first row is equal to zero if $\varphi_h \in \{1, \Pi_h\mathbf{p}, \Pi_h\mathcal{E}(\mathbf{p})\}$. Moreover, since $\{1, \Pi_h\mathbf{p}, \Pi_h\mathcal{E}(\mathbf{p})\} \subset \mathcal{W}_h \cap C^1(\Omega)$, the jump terms in the second row will also vanish. In conclusion, if $\varphi_h \in \{1, \Pi_h\mathbf{p}, \Pi_h\mathcal{E}(\mathbf{p})\}$,

$$
\partial_t \left(f_h, \varphi_h\right)_\Omega = 0.
\tag{31}
$$

□

Remark 4. *The jump terms vanish as long as $\{1, \Pi_h\mathbf{p}, \Pi_h\mathcal{E}(\mathbf{p})\} \subset C^0(\Omega)$, however we require $\Pi_h\mathcal{E}(\mathbf{p}) \in C^1(\Omega)$ to ensure the continuity of the advection field, which means the degree of polynomials k should be at least* 2 *in the test space* (13)*. Of course, there may also be upwind numerical fluxes for discontinuous advection vector field. That is beyond the scope of this paper, and requires further investigation.*

Remark 5. *It can be easily verified that our scheme is still locally conservative, since it can be regarded as traditional LDG applied to advection-diffusion equation with specially designed nonlinear nonlocal coefficients.*

5 Implementation

In this section we will specify the details on the projection operator Π_h and the temporal discretization choice.

5.1 Projection onto the space of continuously differentiable functions

As indicated in Theorem 4, in order to ensure continuity of advection field, the interpolation operator Π_h must project a function onto $\mathcal{W}_h \cap C^1(\Omega)$. To show the existence of such projection, we give an example of Π_h for $\mathbb{Q}^2$ basis functions on N^3 uniform rectangular meshes over the cut-off domain $\Omega = (-L, L)^3$.

Denote the 1D quadratic B-spline basis functions as $\{B_i(p_x)\}$, we look for interpolation in the following form:

$$\Pi_h g(\mathbf{p}) = \sum_{i=1}^{N+2} \sum_{j=1}^{N+2} \sum_{k=1}^{N+2} \alpha_{ijk} B_i(p_x) B_j(p_y) B_k(p_z).$$

Since each 1D basis function is continuously differentiable, the triple product $B_i(p_x)B_j(p_y)B_k(p_z)$ must also be continuously differentiable.

The coefficients α_{ijk} are determined as follows. On the 1D interval $(-L, L)$ with N meshes, define the Greville points as

$$\xi_j = \begin{cases} -L, & j = 1 \\ -L + \frac{2j-3}{2}h, & j = 2, \cdots, N+1 \\ +L, & j = N+2 \end{cases}$$

The coefficients α_{ijk} must be such that the projection coincides with the original function on Greville points, i.e.

$$\sum_{i=1}^{N+2} \sum_{j=1}^{N+2} \sum_{k=1}^{N+2} \alpha_{ijk} B_i(\xi_r) B_j(\xi_s) B_k(\xi_t) = g(\xi_r, \xi_s, \xi_t), \quad \forall\ r, s, t \in \{1, \cdots, N+2\}$$

5.2 Temporal Discretizations

For the sake of efficiency, we tend to use a high-order Runge–Kutta scheme. As we know, RK schemes are a combination of Euler schemes, hence to begin with, we discuss the Euler schemes that preserve conservation laws.

Analogous to [Huang et al.(2023)Huang, Abdelmalik, Breizman and Gamba], we have the following choices:

$$\frac{1}{\Delta t}\left(f_h^{n+1} - f_h^n, \varphi_h\right) = \begin{cases} \mathcal{B}_h\left(f_h^{n+1}, f_h^{n+1}, \varphi_h\right), & \text{fully implicit,} \\ \mathcal{B}_h\left(f_h^{n}, f_h^{n+1}, \varphi_h\right), & \text{semi-implicit,} \\ \mathcal{B}_h\left(f_h^{n+1}, f_h^{n}, \varphi_h\right), & \text{semi-implicit,} \\ \mathcal{B}_h\left(f_h^{n}, f_h^{n}, \varphi_h\right), & \text{fully explicit.} \end{cases} \tag{32}$$

The first two rows are not linear with respect to f^{n+1}, therefore they are too expensive to be solved numerically. The third row seems favorable, however, recall the definition of the semi-discrete weak form in (30); it contains $\left(\nabla_h f_h^n, \mathcal{D}_h[f_h^{n+1}] \cdot \nabla_h \varphi_h\right)$. Therefore, when analyzed as a scheme for an advection-diffusion equation, it is explicit for the diffusion part but as expensive as an implicit scheme. In conclusion, our only choice is the fully explicit scheme.

6 Numerical Experiments

In this section, we assess the proposed structure-preserving LDG method for the relativistic FPL equation in the Maxwell molecule case. This setting retains the tensor structure responsible for relativistic energy conservation while allowing an efficient separable evaluation of the collision operator. We first verify the accuracy of the Lorentz factor projection, showing that the interpolation operator Π_h accurately represents $\mathcal{E}(\mathbf{p}) = \sqrt{1 + |\mathbf{p}|^2}$ and that the corresponding discrete gradient recovers the relativistic velocity field with the expected order of accuracy. We then consider the evolution problem in two regimes: a low-energy regime, where the relativistic equation approaches the non-relativistic FPL equation and can be compared with the BKW reference solution, and a genuinely relativistic regime, where self-convergence is used in the absence of an exact solution. Across these tests, we examine convergence, discrete conservation laws and entropy decay.

6.1 The collision operator for Maxwell molecules

In the collision kernel

$$\Phi(\mathbf{p},\mathbf{q}) = \Lambda\left(\mathbf{p},\mathbf{q}\right)\mathbb{S}\left(\nabla_p\mathcal{E}(\mathbf{p}),\nabla_q\mathcal{E}(\mathbf{q})\right),$$

the Maxwell molecule case corresponds to $\Lambda\left(\mathbf{p},\mathbf{q}\right)\equiv 1$. In this case the relativistic FPL equation reads:

$$\begin{aligned}\partial_t f =& \nabla_p\cdot\left(\int_{\mathbb{R}^3}\mathbb{S}(\mathbf{p},\mathbf{q})\cdot\left(f(\mathbf{q})\nabla_p f(\mathbf{p}) - f(\mathbf{p})\nabla_q f(\mathbf{q})\right)\,d\mathbf{q}\right)\\ =& \nabla_p\cdot\left(\int_{\mathbb{R}^3}\left((\mathbf{u}\cdot\mathbf{u})\mathbb{I}-\mathbf{u}\otimes\mathbf{u}-(\mathbf{z}\times\mathbf{u})\otimes(\mathbf{z}\times\mathbf{u})\right)\cdot\left(f(\mathbf{q})\nabla_p f(\mathbf{p}) - f(\mathbf{p})\nabla_q f(\mathbf{q})\right)\,d\mathbf{q}\right).\end{aligned}$$

where $\mathbf{v}=\nabla_p\mathcal{E}(\mathbf{p})$, $\mathbf{w}=\nabla_q\mathcal{E}(\mathbf{q})$, $\mathbf{u}:=\mathbf{v}-\mathbf{w}$, and $\mathbf{z}:=\frac{\mathbf{v}+\mathbf{w}}{2}$.

Note that in the three terms of tensor field $\mathbb{S}(\mathbf{p},\mathbf{q}) = (\mathbf{u}\cdot\mathbf{u})\mathbb{I}-\mathbf{u}\otimes\mathbf{u}-(\mathbf{z}\times\mathbf{u})\otimes(\mathbf{z}\times\mathbf{u})$, the first and the second terms are of order $\mathcal{O}(|\mathbf{u}|^2)$, while the third term is of order $\mathcal{O}(|\mathbf{z}|^2|\mathbf{u}|^2)$. In the low-energy limit, when almost all the particles have small velocity, say $\mathbf{v},\mathbf{w},\mathbf{u},\mathbf{z}\lesssim\varepsilon\ll 1$, the third term is ε^2 times smaller than the others, hence negligible. In that case we recover the non-relativistic FPL equation.

Another good property of the Maxwell molecule model is that the linear mapping from density f to advection-diffusion fields $\mathcal{U}$ and $\mathcal{D}$ is separable. Indeed, since

$$\begin{aligned}\mathbb{S} =& (\mathbf{u}\cdot\mathbf{u})\mathbb{I}-\mathbf{u}\otimes\mathbf{u}-(\mathbf{z}\times\mathbf{u})\otimes(\mathbf{z}\times\mathbf{u})\\ =& (\mathbf{v}-\mathbf{w})\cdot(\mathbf{v}-\mathbf{w})\mathbb{I}-(\mathbf{v}-\mathbf{w})\otimes(\mathbf{v}-\mathbf{w})-(\mathbf{v}\times\mathbf{w})\otimes(\mathbf{v}\times\mathbf{w}),\end{aligned}$$

if we let

$$\begin{aligned}M &= \int_{\mathbb{R}^3} f(\mathbf{q})d\mathbf{q},\\ W_i &= \int_{\mathbb{R}^3} f(\mathbf{q})w_i d\mathbf{q},\\ X_{ij} &= \int_{\mathbb{R}^3} f(\mathbf{q})w_i w_j d\mathbf{q},\end{aligned}$$

where $w_i=\partial_i\mathcal{E}(\mathbf{q})$, then with Einstein summation notation we have

$$\begin{aligned}\mathcal{D}_{ij}(\mathbf{p}) =& \int_{\mathbb{R}^3}\mathbb{S}_{ij}(\mathbf{p},\mathbf{q})f(\mathbf{q})d\mathbf{q},\\ =& (v_\mu v_\mu M - 2v_\mu W_\mu + X_{\mu\mu})\delta_{ij} - (v_i v_j M - v_i W_j - W_i v_j + X_{ij}) - \varepsilon_{i\xi\eta}\varepsilon_{jst}v_\xi v_s X_{\eta t}.\end{aligned}$$

Analogously, let

$$\begin{aligned}A_\mu &= \int_{\mathbb{R}^3}\partial_\mu f(\mathbf{q})d\mathbf{q}\\ B_{i\mu} &= \int_{\mathbb{R}^3} w_i\partial_\mu f(\mathbf{q})d\mathbf{q}\\ C_{ij\mu} &= \int_{\mathbb{R}^3} w_i w_j\partial_\mu f(\mathbf{q})d\mathbf{q}\end{aligned}$$

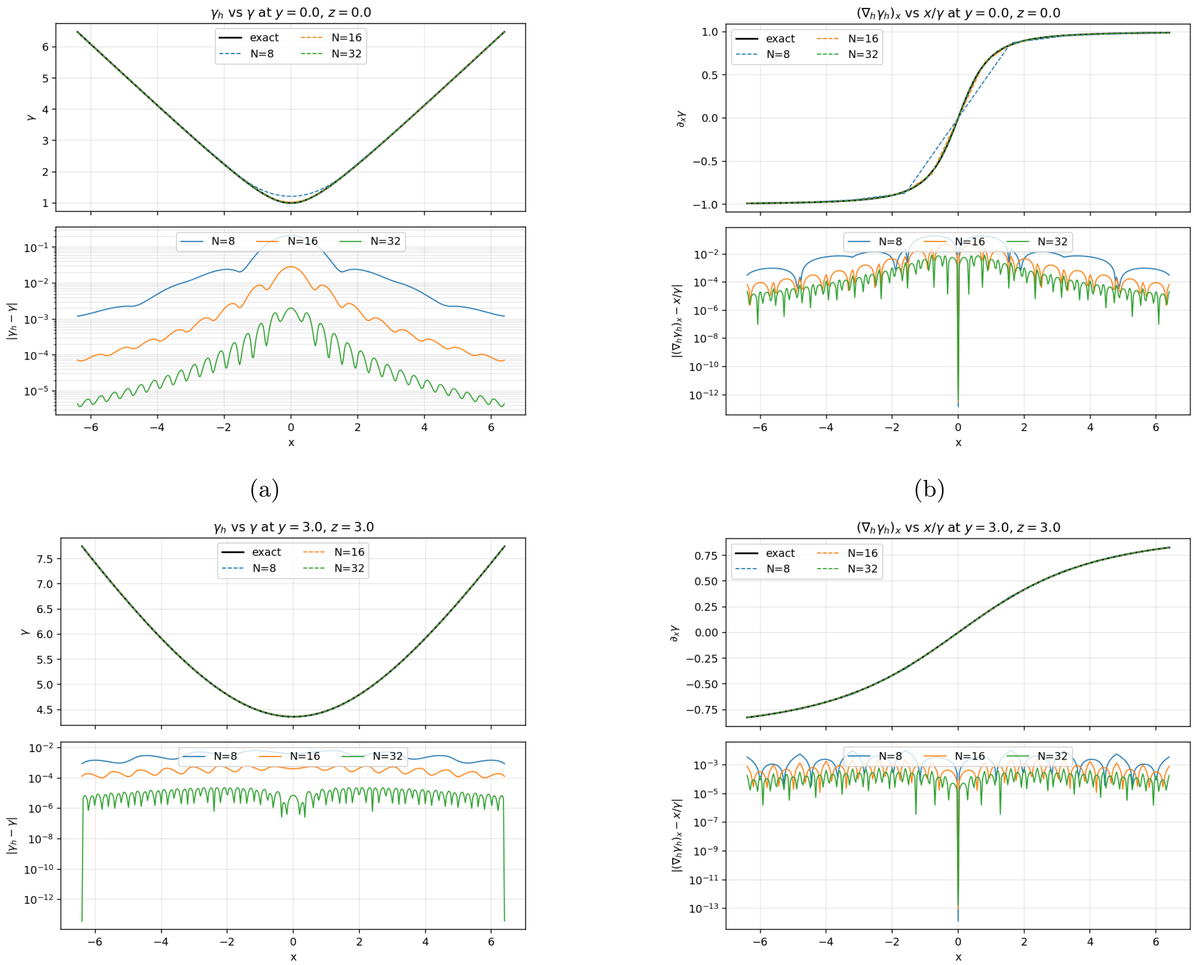
then

$$\begin{aligned}\mathcal{U}_i(\mathbf{p}) =& \int_{\mathbb{R}^3}\mathbb{S}_{ij}(\mathbf{p},\mathbf{q})\partial_j f(\mathbf{q})d\mathbf{q},\\ =& (v_\mu v_\mu A_i - 2v_\mu B_{\mu i} + C_{\mu\mu i}) - (v_i v_j A_j - v_i B_{jj} - B_{ij}v_j + C_{ijj}) - \varepsilon_{i\xi\eta}\varepsilon_{jst}v_\xi v_s C_{\eta tj}.\end{aligned}$$

Suppose that the number of quadrature points and number of basis functions are both of order $\mathcal{O}(N^3)$, the complexity for evaluation of $\mathcal{D}$ and $\mathcal{U}$ is in general $\mathcal{O}(N^6)$, but for Maxwell molecules it can be reduced to $\mathcal{O}(N^3)$ as a consequence of separability.

6.2 Projection of the Lorentz Factor

Define $\gamma(x,y,z) = \sqrt{1+x^2+y^2+z^2}$ on the cut-off domain $\Omega = (-6.4, 6.4)^3$. We apply the interpolation operator Π_h described in 5.1. The result is shown in Figure 1. We expect third order accuracy in $L^2(\Omega)$ and second order accuracy in $H^1(\Omega)$, which is verified in Table 1.



(a) (b) (c) (d)

Figure 1: Slices of the Lorentz factor and its partial derivative at different locations and with various mesh sizes.

N	$\lVert\Pi_h\mathcal{E}(\mathbf{p}) - \mathcal{E}(\mathbf{p})\rVert_{L^2(\Omega)}$	order	$\lVert\nabla_h\Pi_h\mathcal{E}(\mathbf{p}) - \nabla\mathcal{E}(\mathbf{p})\rVert_{L^2(\Omega)}$	order
8	3.09e-1	-	7.06e-1	-
16	2.27e-2	3.8	1.21e-1	2.5
32	2.03e-3	3.5	2.60e-2	2.2

Table 1: Order of accuracy for interpolation operator Π_h.

6.3 BKW solution in the low-energy regime

Note that there exists an exact reference solution to non-relativistic FPL equation for Maxwell molecules, i.e. the BKW solution:

$$f(t, \mathbf{p}) = \frac{1}{(2\pi K)^{3/2}} \exp(-|\mathbf{p}|^2/(2K)) \left(\frac{5K-3}{2K} + \frac{1-K}{2K^2}|\mathbf{p}|^2 \right)$$

where $K = 1 - \exp(-4(t + \frac{1}{4}))$.

As we have mentioned, in the low-energy regime, the relativistic FPL equation is reduced to its non-relativistic counterpart. Note that for Maxwell molecules, if $g(t, \mathbf{p})$ is a solution, then $\alpha^{-3}g(\alpha^{-1}\mathbf{p}, t)$ is also a solution. Therefore we are able to test our method on the following reference solution:

$$f(t, \mathbf{p}) = \frac{1}{(2\pi K)^{3/2}\alpha^3} \exp(-|\mathbf{p}|^2/(2\alpha^2 K)) \left(\frac{5K-3}{2K} + \frac{1-K}{2K^2}|\mathbf{p}|^2/\alpha^2 \right), \tag{33}$$

where $\alpha = 10^{-3}$.

Setting the cut-off domain as $\Omega = (-L, L)^3$ with $L = 6.4 \times 10^{-3}$, using the $\mathbb{Q}^2$ basis, we let the density evolve with the forward Euler method until $t = 0.0112$. The results are shown in Table 2. Since the accuracy is expected to be $\mathcal{O}(h^3 + \Delta t)$, we use one eighth of the previous Δt in each row.

$(N, \Delta t)$	$t = 0.0$	order	$t = 0.0048$	order	$t = 0.0112$	order
(8, 1.6e-4)	5.5e-2	-	5.5e-2	-	5.5e-2	-
(16, 2.0e-5)	1.3e-2	2.1	1.4e-2	2.0	1.3e-2	2.1
(32, 2.5e-6)	1.5e-3	3.1	1.7e-3	3.0	1.6e-3	3.0

Table 2: Order of accuracy in the low-energy regime. We calculate the relative L^2 error by comparing the computed solution with the BKW reference solution.

Fixing $N = 16$ and $\Delta t = 2 \times 10^{-5}$, we let the density evolve with the forward Euler method until $t = 0.4$. In Figure 2, the 1D slices of f_h at different times are plotted. The conservation laws and the entropy decay are shown in Figure 3.

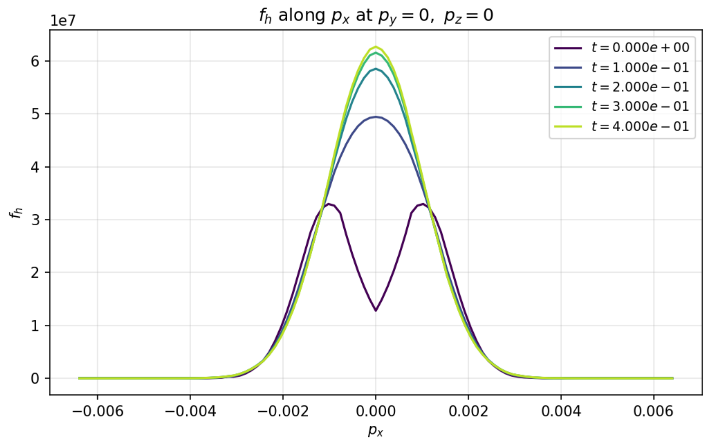
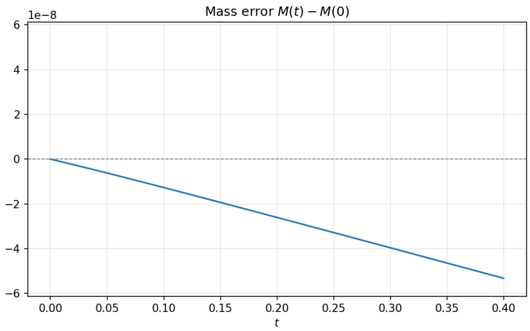
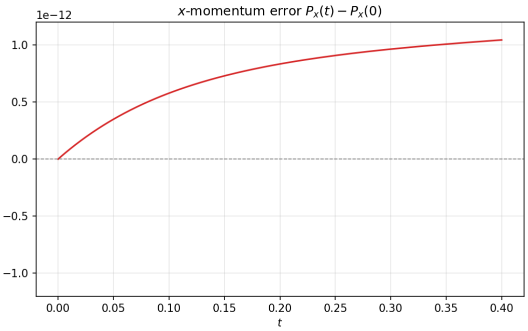


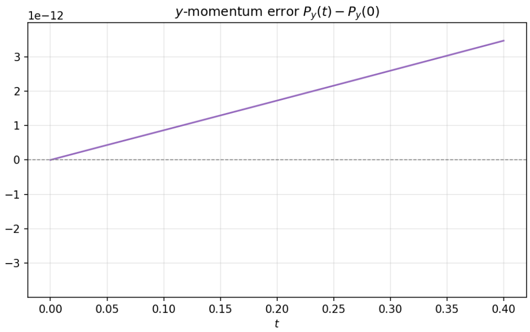
Figure 2: One-dimensional slice of the numerical solution $f_h(p_x, 0, 0, t)$ at different times, with $\alpha = 1.0 \times 10^{-3}$ in BKW solution (33), $N = 16$ and $\Delta t = 2.0 \times 10^{-5}$.



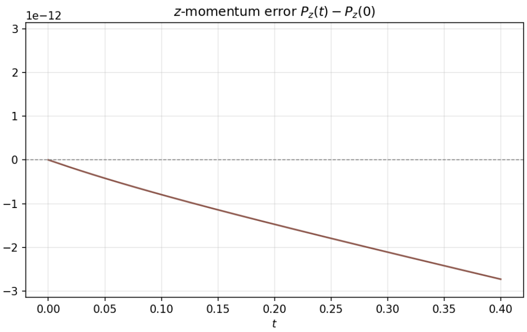
(a) mass error



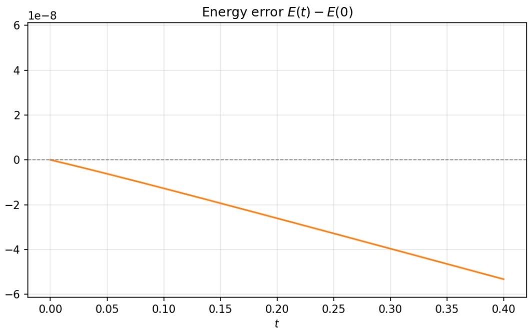
(b) x-momentum error



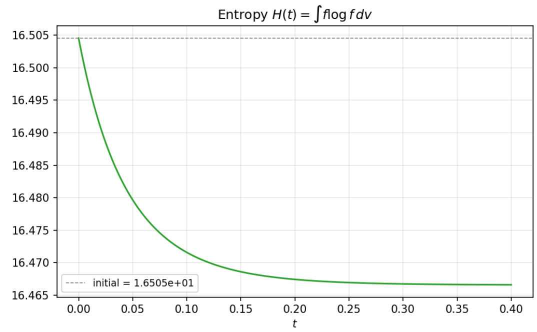
(c) y-momentum error



(d) z-momentum error



(e) energy error



(f) entropy decay

Figure 3: Conservation laws and entropy decay in the low-energy case, with $\alpha = 1.0 \times 10^{-3}$ in BKW solution (33), $N = 16$ and $\Delta t = 2.0 \times 10^{-5}$.

Note that in this test case, the mass and energy errors are much higher than machine epsilon. This is not due to the scheme but a consequence of amplified rounding error. When the width $\alpha = 1.0 \times 10^{-3}$, f_h is of the order 10^7 and its gradient $\nabla_h f_h$ is at least of order 10^{10}, then by looking at the integral in the right hand side of (30) large values of f_h and $\nabla_h f_h$ will amplify the term $\Phi_h(\mathbf{p}, \mathbf{q}) \cdot (\nabla_{h,q} \varphi_h(\mathbf{q}) - \nabla_{h,p} \varphi_h(\mathbf{p}))$ even though it is machine zero. To validate our claim, we perform another test with $\alpha = 0.1$ for comparison. The results are shown in Figure 4, and it can be observed that the conservation law is preserved up to machine epsilon in this case.

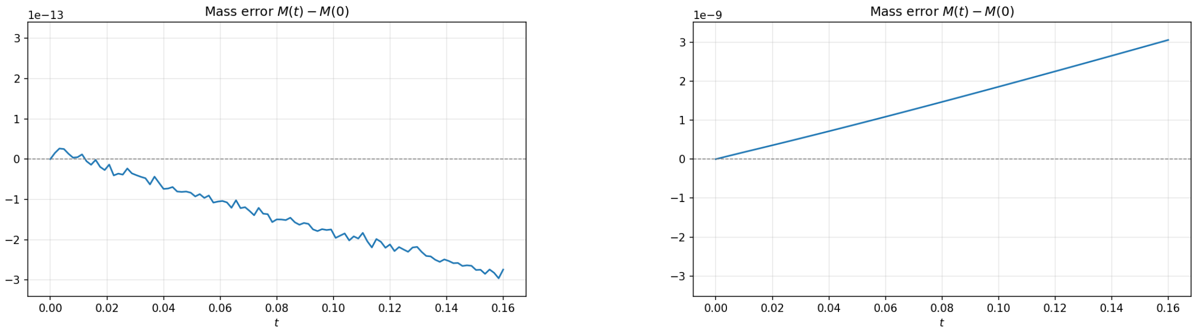


Figure 4: Comparison of the conservation error with different initial conditions, in both cases $N = 8$ and $\Delta t = 1.6 \times 10^{-4}$. Left: initial width $\alpha = 0.1$ and cut-off domain $\Omega = (-6.4 \times 10^{-1}, 6.4 \times 10^{-1})^3$; Right: initial width $\alpha = 0.001$ and cut-off domain $\Omega = (-6.4 \times 10^{-3}, 6.4 \times 10^{-3})^3$. The parameter α controls the width and amplitude of solutions through Equation 33.

6.4 Self-convergence test in relativistic regime

In this subsection we focus on the relativistic regime. We set the cut-off domain as $\Omega = (-6.4, 6.4)^3$, use $N = 8, 16, 32$ uniform meshes in each dimension, and use the $\mathbb{Q}^2$ basis. The initial condition is as follows:

$$f_0(\mathbf{p}) = \exp\left(-\frac{|\mathbf{p} - \mathbf{p}_1|^2}{2}\right) + \exp\left(-\frac{|\mathbf{p} - \mathbf{p}_2|^2}{2}\right),$$

with $\mathbf{p}_1 = (-2, 0, 0)$ and $\mathbf{p}_2 = (2, 0, 0)$.

In the relativistic regime, there is no exact reference solution. Therefore, we perform the self-convergence test; the results are shown in Table 3. Note that we choose a small enough step size $\Delta t = 1.0 \times 10^{-5}$ so that the error from time discretization is much smaller than that from spatial discretization.

Fixing $N = 16$ and $\Delta t = 1 \times 10^{-5}$, we let the density evolve with the forward Euler method until $t = 0.2$. In Figure 5, the 2D slices of f_h at different times are plotted. The conservation laws and the entropy decay are shown in Figure 6.

$(N, \Delta t)$	$t = 0.0$	order	$t = 0.01$	order	$t = 0.02$	order
(8, 1.0e-5)	1.2e-1	-	1.1e-1	-	1.1e-1	-
(16, 1.0e-5)	1.3e-2	3.2	1.1e-2	3.3	1.0e-2	3.5
(32, 1.0e-5)	reference	-	reference	-	reference	-

Table 3: Order of accuracy in the high-energy regime. We calculate the L^2 error by comparing with the computed solution on the finest grid, shown in the third row.

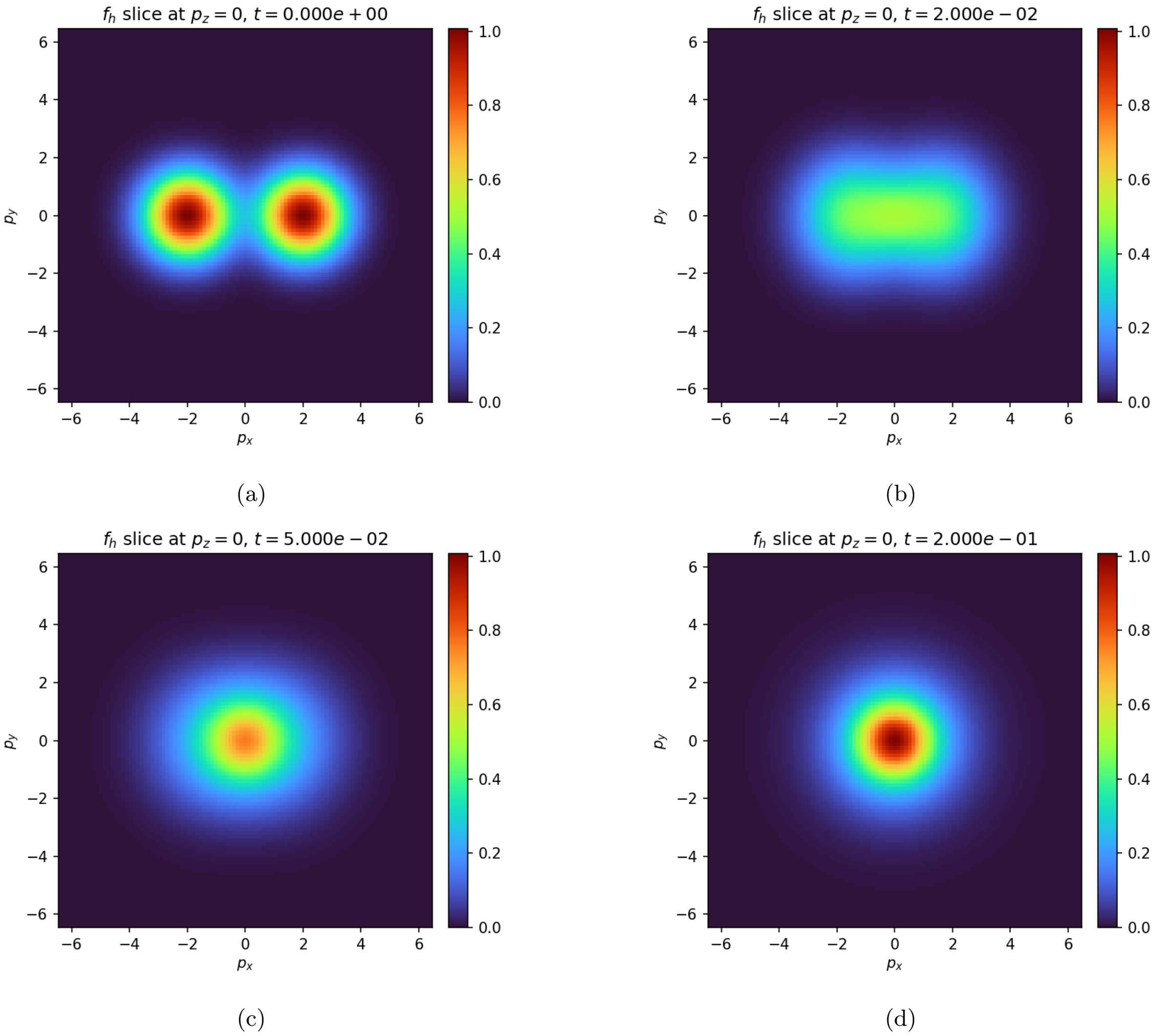
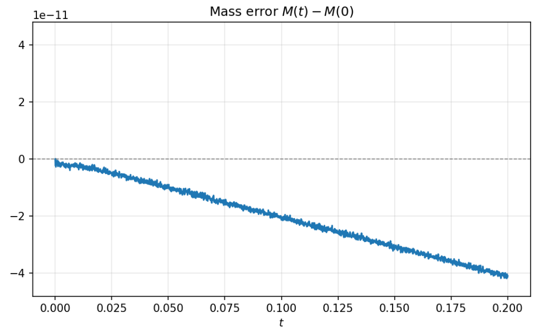
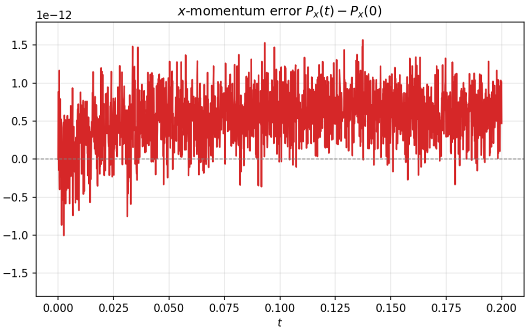


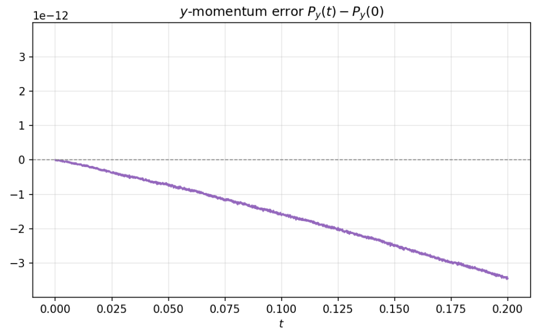
Figure 5: Two-dimensional slices of the numerical solution $f_h(p_x, p_y, 0, t)$ at different times, with cut-off domain $\Omega = (-6.4, 6.4)^3$, $N = 16$ and $\Delta t = 1.0 \times 10^{-5}$.



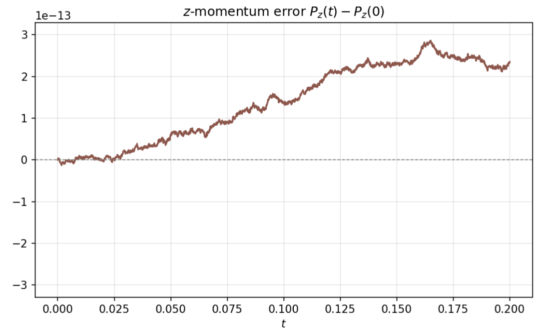
(a) mass error



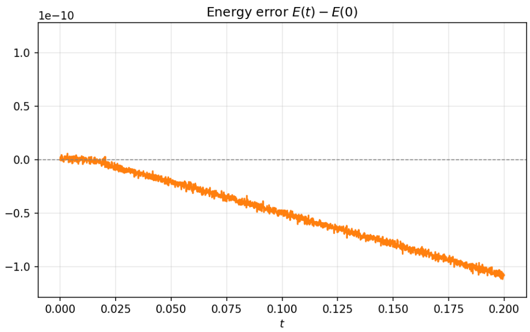
(b) x-momentum error



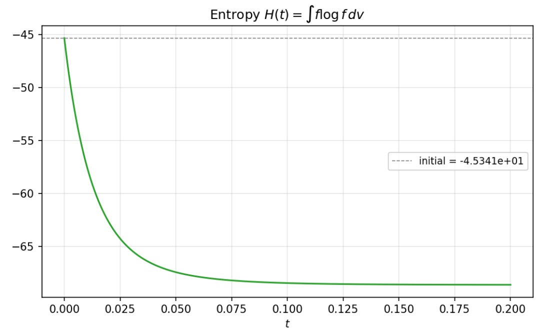
(c) y-momentum error



(d) z-momentum error



(e) energy error



(f) entropy decay

Figure 6: Conservation law and entropy decay, with cut-off domain $\Omega = (-6.4, 6.4)^3$, $N = 16$ and $\Delta t = 1.0 \times 10^{-5}$.

7 Concluding Remarks

In this work, we developed a structure-preserving local discontinuous Galerkin (LDG) method for the Fokker–Planck–Landau (FPL) equation. By introducing a discrete gradient operator and carefully discretizing the collision kernel using projections of the particle energy, we constructed a scheme that simultaneously satisfies discrete conservation of mass, momentum, and energy. Unlike previous conservative methods, our approach ensures that the underlying geometric structure of the collision operator—specifically the null space of the tensor kernel—is preserved at the discrete level.

To resolve the contradiction between stability and conservation, we proposed an upwind structure-preserving scheme that exploits the unique treatment of jump terms in the LDG framework. The resulting method exhibits enhanced L^2 stability and is amenable to high-order discretizations and parallel implementation.

We validated our approach through numerical experiments on the relativistic FPL equation. By testing the relativistic solver in both low-energy and high-energy regimes, the order of accuracy is verified. Moreover, the conservation laws are preserved to nearly machine precision.

In future work, we plan to apply this framework to the relativistic FPL equation with Coulomb potential, through potentially low-rank decomposition of the collision kernel. Moreover, we will explore adaptive mesh refinement, and investigate coupling with Vlasov solvers in high-dimensional phase space.

8 Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

9 Acknowledgements

The authors thank and gratefully acknowledge the support from the Oden Institute of Computational Engineering and Sciences, the Institute for Fusion Studies and the University of Texas Austin. The authors acknowledge the Texas Advanced Computing Center (TACC) at The University of Texas at Austin for providing computational resources that have contributed to the research results reported within this paper.

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